

Perturbation evidence is multiplying faster than it can be organised

A genome-scale Perturb-Seq study can profile **more than 2.5 million cells**, and pooled screens can test thousands of targets in one experiment. That scale promises the statistical power to distinguish robust biology from study-specific noise—but the evidence is split across repositories, identifiers, assay conventions, reference genomes and incompatible analysis choices. Without a shared account of what was perturbed, in which model, with which readout and under which pipeline, combining studies remains slow, fragile and difficult to reproduce. The Catalogue makes that cross-study power practical: one curated schema, one stable target identity and one access path for comparing results.

36× more papers per year than in 2015



Europe PMC title/abstract query; 24 Aug 2026. Scale example: Replogle et al., Cell 2022.

A single catalogue turns scattered evidence into a searchable resource

1,237

DATASETS

1,201 CRISPR screens

gene perturbation × phenotype or fitness

26 Perturb-Seq datasets

perturbation × single-cell transcriptome

10 MAVE datasets

variant × functional measurement

19,235

human targets

36

tissues

34

cell types

1,200

cell lines

Metadata and reanalysis make results comparable

84

searchable fields in the dataset contract

Study & provenance

Title, authors, year, source URI, licence, linked datasets

Model system

Organism, tissue, cell type or line, disease, sex, stage

Perturbation & library

Targets, library design, delivery, guides, variants

Readout & processing

Assay, technology, sequencing kit, reference, software

Perturb-Seq reprocessing starts again from raw reads

Author matrices differ in alignment, annotation, filtering and guide assignment. Selected studies follow one shared computational path.

SRA / FASTQ

Raw sequencing reads and study-specific probe lists

→

Cell × gene counts

Shared reference and gene annotation

→

Guide assignment

Map cells to perturbations

→

QC + filtering

Produce the final analysis-ready H5AD

Perturbation vs control

→

DEA

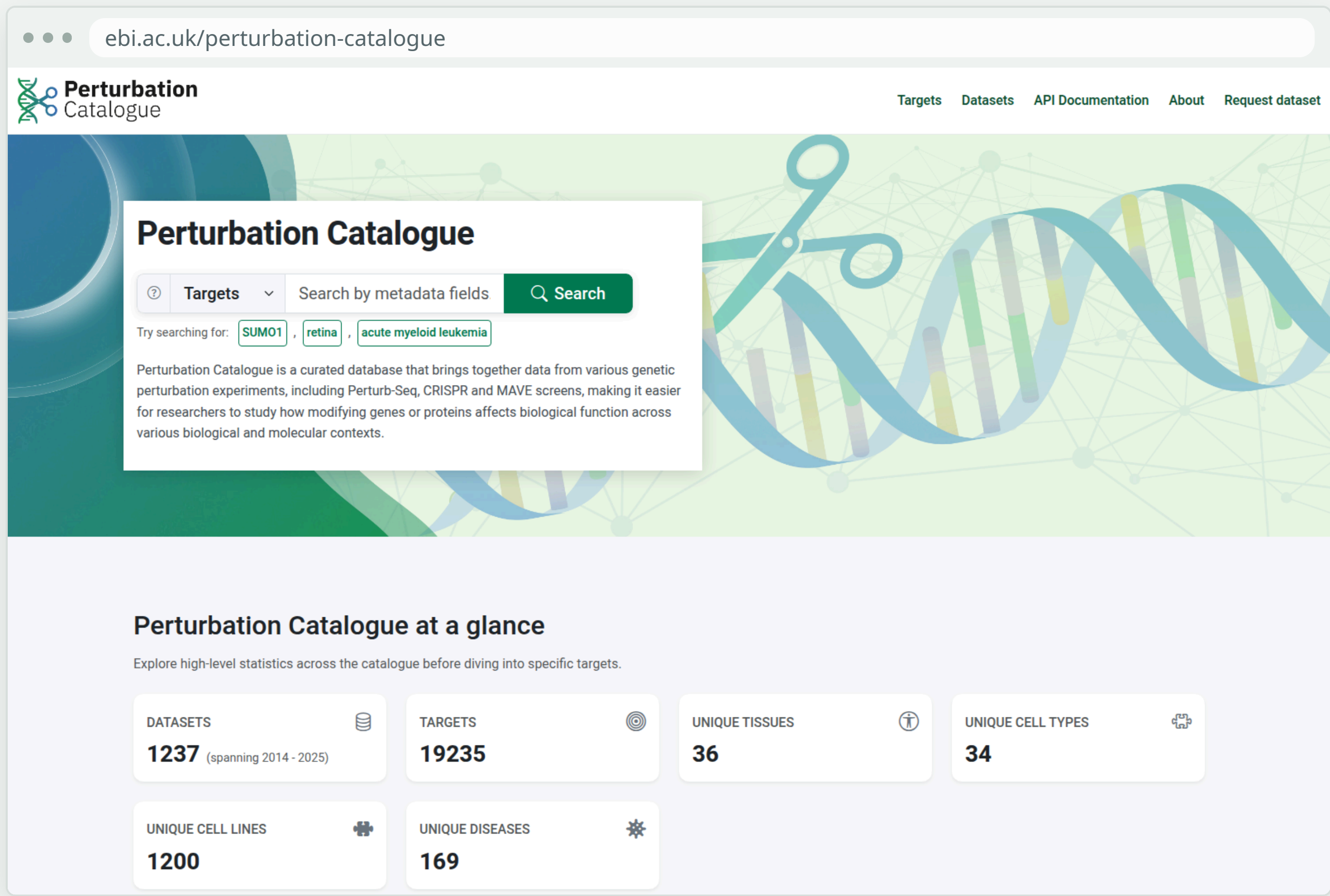
→

Hallmark GSEA

Five datasets are live. Broad agreement with author-generated matrices is checked where available; independent quality metrics and wider kit support are next.

Stable Ensembl gene IDs and ontology-linked entities make the

Explore the Catalogue in a browser



Target search, current coverage and biological-context summaries in one human-facing portal.

Every result is programmatically accessible

Search, metadata, modality results and complete downloads use the same production backend as the portal.

```
GET /search?query=BRCA2&search_mode=targets
GET /dataset/nadig_2025_jurkat
GET /v1/perturb-seq/nadig_2025_jurkat/search?limit=1
GET /v1/perturb-seq/nadig_2025_jurkat/download?format=csv.gz
```

```
{
  "total_rows_count": 30078802,
  "results": [{
    "perturbation": {"gene_symbol": "TFAM"},
    "effect": {
      "gene_symbol": "MT-ND5",
      "log2fc": -1.926, "padj": 0.0
    }
  }]
}
```

Next, integration becomes continuous and the analytical foundation becomes stronger

Extract structured metadata faster while retaining curator review

AI-assisted drafting, ambiguity resolution and validated ontology mapping before ingestion.

Reanalyse more designs and measure quality independently

More kits and experimental structures, author-independent QC and modality-appropriate methods.

Publish periodic, versioned releases with explicit provenance

Stable snapshots and transparent analytical changes for reproducible downstream use.

Bring us the next hard perturbation-data problem

Suggest datasets, challenge the schema or develop the resource with us. We welcome scientific partnerships and funded multi-year collaborations around your programme's modalities, analyses and functionality.

perturbation-catalogue-help@ebi.ac.uk

Explore the Catalogue



Industry and research collaboration

